Car Accidents - Likelihood

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Data Science Capstone 2

Overview

- Dramaticized?
- Newer = Better?
- Does it matter?

The Affected Audience?

- Insurance Providers / Actuaries

- Concerned Citizens / Drivers

Factors & Data Info

- Source: CA Crash Dataset

- ~1,000,000 rows across 2 datasets and 50 features

Data Structure

	_id	PartyId	CollisionId	PartyNumber	PartyType	IsAtFault	IsOnDutyEmergencyVehicle	IsHitAndRun	AirbagCode	AirbagDescription	...	Vehicle2TypeDesc	Vehicle2Year	Vehicle2Make	Vehicle2Model	Vehicle2C
0	1	13900346	4541902	1	Driver	True	False	True	B	UNKNOWN	...	NaN	NaN	NaN	NaN	N
1	2	13900345	4541901	2	Driver	NaN	False	False	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
2	3	13900344	4541901	1	Driver	True	False	False	L	AIR BAG DEPLOYED	...	NaN	NaN	NaN	NaN	N
3	4	13900343	4541900	2	Driver	NaN	False	False	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
4	5	13900342	4541900	1	Driver	NaN	False	False	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
767047	767048	14979609	5105674	1	Driver	True	False	NaN	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
767048	767049	14979608	5105673	1	Driver	True	False	NaN	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
767049	767050	14979607	5105672	2	Driver	NaN	False	NaN	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
767050	767051	14979606	5105672	1	Driver	True	False	NaN	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N
767051	767052	14979605	5105671	2	Driver	NaN	False	NaN	M	AIR BAG NOT DEPLOYED	...	NaN	NaN	NaN	NaN	N

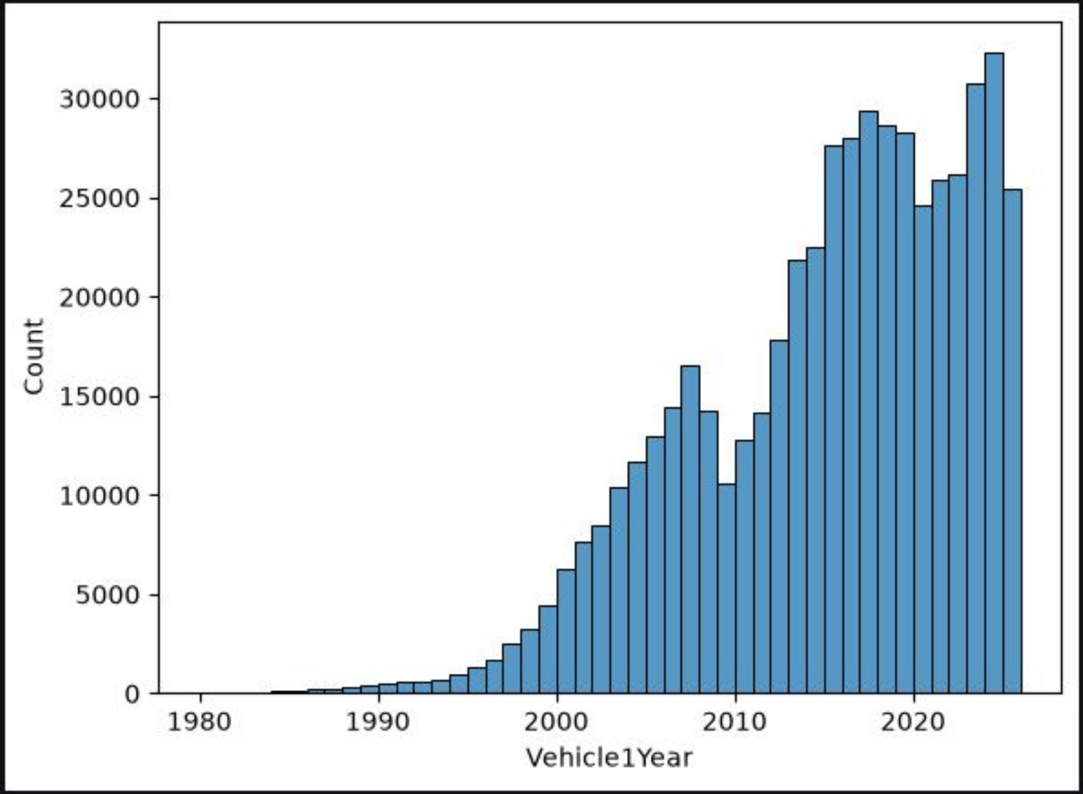
767052 rows × 49 columns

Data Structure

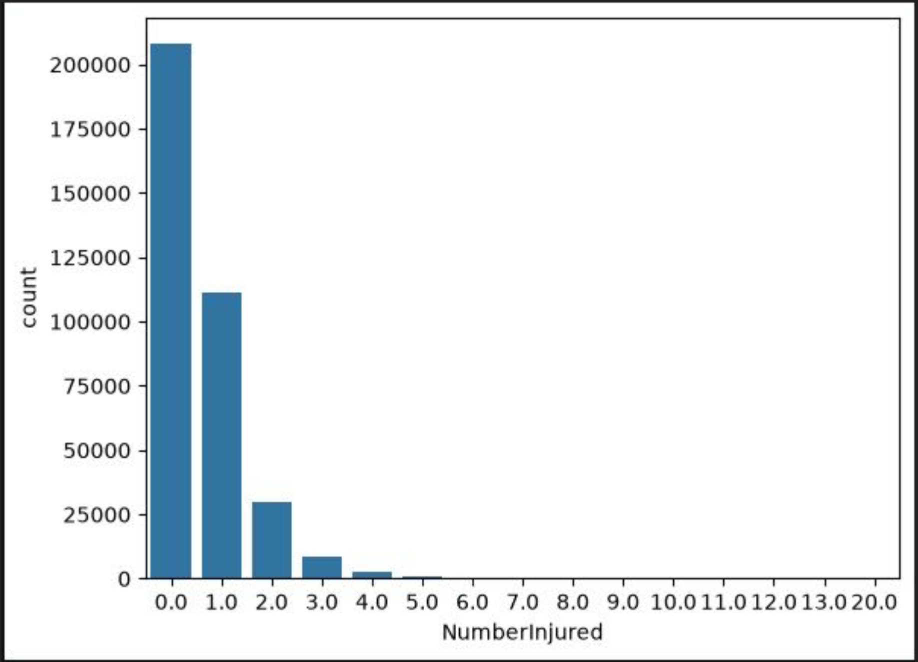
	Collision Id	IsTowAway	MotorVehicleInvolvedWithDesc	MotorVehicleInvolvedWithOtherDesc	NumberInjured	NumberKilled	Primary Collision Factor Code	Primary Collision Factor Violation	PrimaryCollisionFactorIsCited	PrimaryCollisionPartyNumber
0	4550265	True	NON-COLLISION	NaN	1.0	0.0	A	VC 22107	False	1.0
1	4550264	False	OTHER MOTOR VEHICLE	NaN	1.0	0.0	A	22350 VC	False	1.0
2	4550263	False	OTHER MOTOR VEHICLE	NaN	0.0	0.0	A	VC 22107	False	1.0
3	4550262	False	OTHER OBJECT	METAL OBJECT	0.0	0.0	A	VC 22350	False	1.0
4	4550261	True	OTHER MOTOR VEHICLE	NaN	3.0	0.0	A	VC 22450(a)	False	1.0
...	...	...	...	...	...	...	...	...	...	...
393505	5105490	True	OTHER MOTOR VEHICLE	NaN	0.0	0.0	A	21801(a)	False	1.0
393506	5105489	True	OTHER MOTOR VEHICLE	NaN	2.0	0.0	A	21453(a)	False	1.0
393507	5105488	False	OTHER MOTOR VEHICLE	NaN	1.0	0.0	A	21801(a)	False	1.0
393508	5105487	True	OTHER MOTOR VEHICLE	NaN	1.0	0.0	A	22350	False	1.0
393509	5105486	True	FIXED OBJECT	CURB	0.0	0.0	A	22350	False	1.0

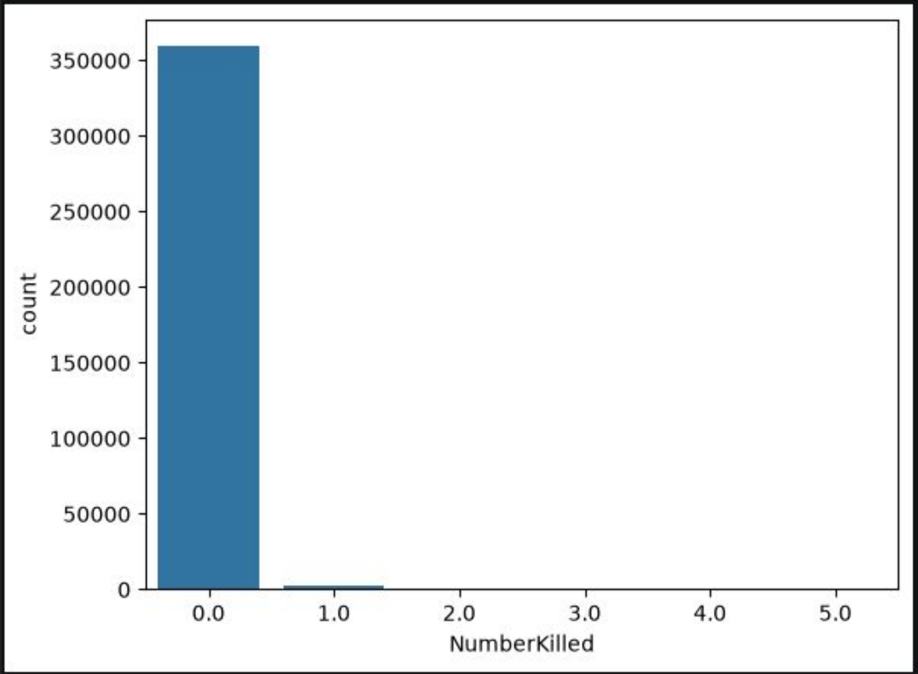
393510 rows × 10 columns

Real World Statistics



Real World Statistics





Model Creation

- ~360,000 rows across 20 Features to determine likelihood

- Model types used:
  - Gini (Random Forests)
  - Entropy (Random Forests)
  - Linear Regression

Model Creation

```
model_pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('forest', RandomForestClassifier(n_estimators = 125, criterion = 'entropy', max_depth= 15 ,random_state = 42, min_samples_split= 6))
])

model_pipeline.fit(X_train, y_train)
train_accuracy = model_pipeline.score(X_train,y_train)
enthropy_predict = model_pipeline.predict(X_test)
accuracy = model_pipeline.score(X_test,y_test)
print(train_accuracy)
print(accuracy)
print(enthropy_predict)
```

```
0.6011408853519025
0.59248727654719
```

Model Creation

```
model_pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('forest', RandomForestClassifier(bootstrap=True, n_estimators = 125, criterion = 'gini', max_depth= 15, random_state = 42, min_samples_split= 6))
])

model_pipeline.fit(X_train, y_train)
gini_predict = model_pipeline.predict(X_test)
accuracy = model_pipeline.score(X_test,y_test)

print(accuracy)
```

```
0.5924148298407984
```

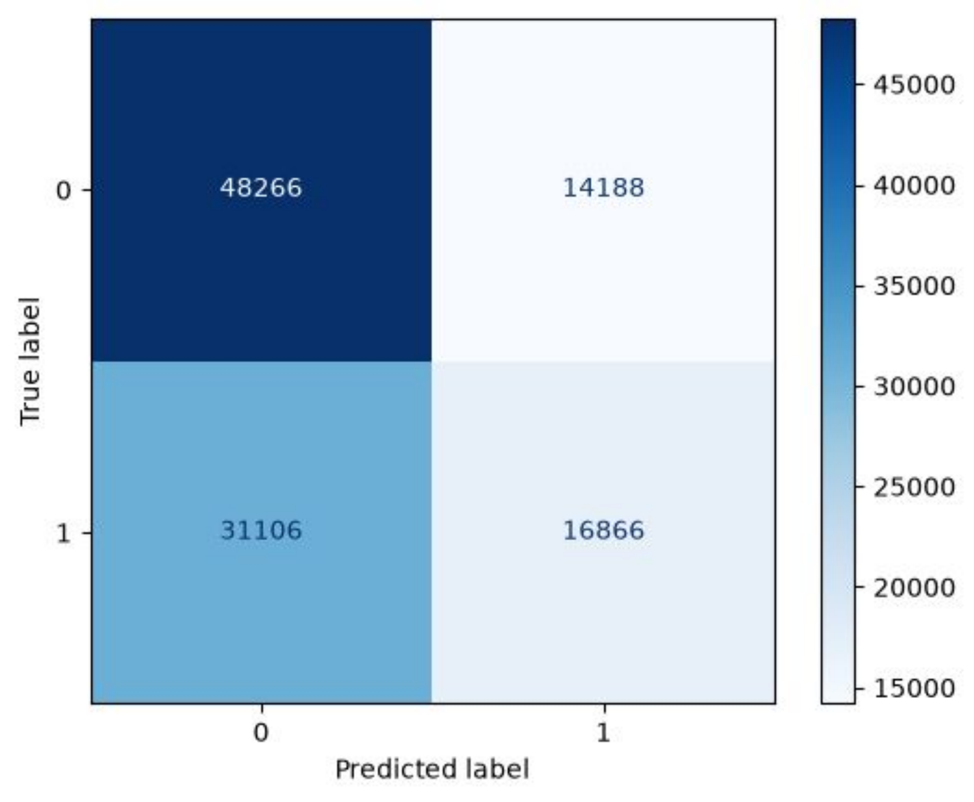
Model Creation

```
model_pipeline = Pipeline([
    ('preprocessor', preprocessor),
    ('LogisticRegressor', LogisticRegression(C = 1, solver = 'saga', random_state= 42))
])

model_pipeline.fit(X_train, y_train)
train_accuracy = model_pipeline.score(X_train,y_train)
accuracy = model_pipeline.score(X_test,y_test)
print(train_accuracy)
print(accuracy)
```

```
c:\Python312\Lib\site-packages\sklearn\linear_model\_sag.py:348: ConvergenceWarning: The max_iter was reached which means the coef_ did not converge
  warnings.warn(
0.591939194230291
0.5898248600872983
```

Model Results



Ways To Improve

- Improve Data Quality
- Change the Modeling Process / Parameters
- Explore Different Models

Conclusion

More Work Needs to Be Done